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     ****************************************************
     *                                                  *
     *           STAAD.Pro V8i SELECTseries2            *
     *           Version  20.07.07.32                   *
     *           Proprietary Program of                 *
     *           Bentley Systems, Inc.                  *
     *           Date=    JAN 16, 2012                  *
     *           Time=    13:47:28                      *
     *                                                  *
     *      USER ID: RNS _ASSOCIATES                    *
     ****************************************************


     1. STAAD SPACE DXF IMPORT OF 20111202 JRC PIG AREA STAIRCASE.DXF
INPUT FILE: 20111216 STAIRCASE 2 PIG STATION.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 03-DEC-11
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 0 0 1; 3 0 1.3065 0; 4 0 1.3065 1; 5 0 2.613 0; 6 0 2.613 1
     9. 7 0 4.0065 0; 8 0 4.0065 1; 9 0 5.4 0; 10 0 5.4 1; 11 0.5 5.4 0; 12 0.5 5.4 1
    10. 13 1 5.4 0; 14 1 5.4 1; 15 1.5 0 0; 16 1.5 0 1; 17 1.5 1.3065 0
    11. 18 1.5 1.3065 1; 19 1.5 2.613 0; 20 1.5 2.613 1; 21 1.5 4.0065 0
    12. 22 1.5 4.0065 1; 23 1.5 5.4 0; 24 1.5 5.4 1; 25 5.6 4E-016 0; 26 5.6 4E-016 1
    13. 29 5.1 2.613 0; 30 5.1 2.613 1; 31 6.1 2.613 0; 32 6.1 2.613 1
    14. 33 6.6 4E-016 0; 34 6.6 4E-016 1; 37 7.1 2.613 0; 38 7.1 2.613 1
    15. 39 10.475 4E-016 0; 40 10.475 4E-016 1; 41 5.6 1.3065 0; 42 5.6 1.3065 1
    16. 43 5.6 2.613 0; 44 5.6 2.613 1; 45 6.6 1.3065 0; 46 6.6 1.3065 1
    17. 47 6.6 2.613 0; 48 6.6 2.613 1
    18. MEMBER INCIDENCES
    19. 1 1 3; 2 3 2; 3 4 1; 4 2 4; 5 4 3; 6 3 5; 7 5 4; 8 6 3; 9 4 6; 10 5 6; 11 5 7
    20. 12 7 6; 13 8 5; 14 6 8; 15 8 7; 16 7 9; 17 9 8; 18 10 7; 19 8 10; 20 9 10
    21. 21 9 11; 22 10 12; 23 11 12; 24 3 17; 25 4 18; 26 5 19; 27 6 20; 28 7 21
    22. 29 8 22; 30 11 13; 31 12 14; 32 13 14; 33 13 23; 34 14 24; 35 17 15; 36 17 16
    23. 37 18 15; 38 18 16; 39 18 17; 40 19 17; 41 19 18; 42 20 17; 43 20 18; 44 19 20
    24. 45 21 19; 46 21 20; 47 22 19; 48 22 20; 49 22 21; 50 23 21; 51 23 22; 52 24 21
    25. 53 24 22; 54 23 24; 55 23 29; 56 24 30; 57 41 25; 59 42 25; 60 42 26; 61 42 41
    26. 62 43 41; 63 43 42; 64 44 41; 65 44 42; 66 29 30; 67 29 43; 68 30 44; 71 31 32
    27. 72 31 47; 73 32 48; 74 45 33; 75 45 34; 76 46 33; 77 46 34; 78 46 45; 79 47 45
    28. 80 47 46; 81 48 45; 82 48 46; 83 37 38; 84 37 39; 85 38 40; 86 41 45; 87 42 46
    29. 88 43 31; 89 44 32; 92 47 37; 93 48 38; 94 41 26; 95 43 44; 96 47 48
    30. DEFINE MATERIAL START
    31. ISOTROPIC STEEL
    32. E 2.05E+008
    33. POISSON 0.3
    34. DENSITY 76.8195
    35. ALPHA 1.2E-005
    36. DAMP 0.03
    37. END DEFINE MATERIAL
    38. MEMBER PROPERTY JAPANESE
    39. 1 4 6 9 11 14 16 19 35 38 40 43 45 48 50 53 57 60 62 65 74 77 79 -
    40. 82 TABLE ST H150X150X7
    41. 5 10 15 20 39 44 49 54 61 78 95 96 TABLE ST H150X150X7
    42. 55 56 67 68 84 85 92 93 TABLE ST C200X80X7.5
    43. 23 32 66 71 83 TABLE ST C100X50X5
    44. 2 3 7 8 12 13 17 18 36 37 41 42 46 47 51 52 59 63 64 75 76 80 81 -
    45. 94 TABLE ST C100X50X5
    46. 21 22 24 TO 31 33 34 72 73 86 TO 89 TABLE ST H150X150X7
    47. CONSTANTS
    48. BETA 180 MEMB 55 67 84 92
    49. MATERIAL STEEL ALL
    50. SUPPORTS
    51. 1 2 15 16 25 26 33 34 PINNED
    52. 39 40 FIXED BUT MY MZ KMX 0.00099
    53. MEMBER RELEASE
    54. 5 10 15 20 39 44 49 54 61 66 78 83 START MY MZ
    55. 5 10 15 20 39 44 49 54 61 66 78 83 END MY MZ
    56. 55 56 START MY MZ
    57. 23 32 71 START MY MZ
    58. 23 32 71 END MY MZ
    59. 95 START MY MZ
    60. 95 END MY MZ
    61. 96 START MY MZ
    62. 96 END MY MZ
    63. 67 END MY MZ
    64. 68 END MY MZ
    65. 93 START MY MZ
    66. 92 END MY MZ
    67. MEMBER TENSION
    68. 2 3 7 8 12 13 17 18 36 37 41 42 46 47 51 52 59 63 64 75 76 80 81
    69. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD: DL
    70. SELFWEIGHT Y -1
    71. MEMBER LOAD
    72. *CHEQUERED PLATE LOAD 0.5KN/M^2 X 0.5M = 0.25 KN/M
    73. 55 56 84 85 UNI GY -0.25
    74. *CHEQUERED PLATE LOAD 0.5KN/M^2 X 0.5M = 0.25 KN/M
    75. 23 32 71 95 96 UNI GY -0.25
    76. *CHEQUERED PLATE LOAD 0.5KN/M^2 X 0.25M = 0.125 KN/M
    77. 20 54 66 83 UNI GY -0.125
    78. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD: LL
    79. MEMBER LOAD
    80. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M
    81. 55 56 84 85 UNI GY -2.5
    82. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M
    83. 23 32 71 95 96 UNI GY -2.5
    84. *LIVE LOAD 5KN/M^2 X 0.25M = 1.25 KN/M
    85. 20 54 66 83 UNI GY -1.25
    86. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD X: WLX
    87. *WIND LOAD 0.85KN/M^2 X 0.15M = 2.5 KN/M X 0.0120 = 0.017KN/M
    88. MEMBER LOAD
    89. 1 4 TO 6 9 TO 11 14 TO 16 19 20 35 38 TO 40 43 TO 45 48 TO 50 53 TO 57 60 -
    90. 61 TO 62 65 66 71 74 77 TO 79 82 TO 85 96 UNI GX -0.17
    91. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD Z: WLZ
    92. *WIND LOAD 0.85KN/M^2 X 0.15M = 2.5 KN/M X 0.0120 = 0.017KN/M
    93. MEMBER LOAD
    94. 1 4 6 9 11 14 16 19 21 22 24 TO 31 33 TO 35 38 43 45 48 50 53 55 TO 57 60 -
    95. 62 65 67 68 72 TO 74 77 79 82 84 TO 89 92 93 UNI GZ -0.17
    96. LOAD 5 LOADTYPE NONE  TITLE NOTIONAL LOAD X: NLX
    97. MEMBER LOAD
    98. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M X 0.015 = 0.0375KN/M
    99. 55 56 84 85 UNI GX -0.0375
   100. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M X 0.015 = 0.0375KN/M
   101. 23 32 71 95 96 UNI GX -0.0375
   102. *LIVE LOAD 5KN/M^2 X 0.25M = 1.25 KN/M X 0.015 = 0.0188KN/M
   103. 20 54 66 83 UNI GX -0.0188
   104. LOAD 6 LOADTYPE NONE  TITLE NOTIONAL LOAD Z: NLZ
   105. MEMBER LOAD
   106. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M X 0.015 = 0.0375KN/M
   107. 55 56 84 85 UNI GZ -0.0375
   108. *LIVE LOAD 5KN/M^2 X 0.5M = 2.5 KN/M X 0.015 = 0.0375KN/M
   109. 23 32 71 95 96 UNI GZ -0.0375
   110. *LIVE LOAD 5KN/M^2 X 0.25M = 1.25 KN/M X 0.015 = 0.0188KN/M
   111. 20 54 66 83 UNI GZ -0.0188
   112. LOAD COMB 7 1.0DL + 1.0LL
   113. 1 1.0 2 1.0
   114. LOAD COMB 8 1.0DL + 1.0LL +1.0WLX
   115. 1 1.0 2 1.0 3 1.0
   116. LOAD COMB 9 1.0DL +1.0LL + 1.0WLZ
   117. 1 1.0 2 1.0 4 1.0
   118. LOAD COMB 10 1.0DL + 1.0LL +1.0NLX
   119. 1 1.0 2 1.0 5 1.0
   120. LOAD COMB 11 1.0DL +1.0LL + 1.0NLZ
   121. 1 1.0 2 1.0 6 1.0
   122. LOAD COMB 12 1.4DL +1.6LL
   123. 1 1.4 2 1.6
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124. LOAD COMB 13 1.2DL + 1.2LL +1.2WLX
125. 1 1.2 2 1.2 3 1.2
126. LOAD COMB 14 1.2DL +1.2LL + 1.2WLZ
127. 1 1.2 2 1.2 4 1.2
128. LOAD COMB 15 1.2DL + 1.2LL +1.2NLX
129. 1 1.2 2 1.2 5 1.2
130. LOAD COMB 16 1.2DL +1.2LL + 1.2NLZ
131. 1 1.2 2 1.2 6 1.2
132. PERFORM ANALYSIS
```

P R O B L E M   S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =   44/   91/   10

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH=   14/   8/   48 DOF
TOTAL PRIMARY LOAD CASES =   6, TOTAL DEGREES OF FREEDOM =   234
SIZE OF STIFFNESS MATRIX =   12 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE =   12.3/ 366823.8 MB

```
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=    1

**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=    2

**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=    3

**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=    4

**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=    5

**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=    6
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133. LOAD LIST 7 TO 11
134. PRINT JOINT DISPLACEMENTS ALL
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JOINT DISPLACEMENT (CM   RADIANS)   STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
1	7	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0003
	8	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001
	9	0.0000	0.0000	0.0000	-0.0001	0.0000	-0.0003
	10	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0003
	11	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0003
2	7	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001
	8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002
	9	0.0000	0.0000	0.0000	-0.0001	0.0000	-0.0001
	10	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001
	11	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001
3	7	0.0320	-0.0002	0.0004	0.0000	0.0000	-0.0001
	8	0.0102	-0.0002	0.0005	0.0000	0.0000	-0.0001
	9	0.0302	-0.0016	-0.0064	0.0000	0.0000	-0.0001
	10	0.0315	-0.0002	0.0004	0.0000	0.0000	-0.0001
	11	0.0317	-0.0002	0.0002	0.0000	0.0000	-0.0001
4	7	0.0085	-0.0007	0.0004	0.0000	0.0000	0.0000
	8	-0.0159	-0.0009	0.0005	0.0000	0.0000	0.0000
	9	0.0094	0.0001	-0.0067	-0.0001	0.0000	0.0000
	10	0.0077	-0.0008	0.0004	0.0000	0.0000	0.0000
	11	0.0087	-0.0007	0.0002	0.0000	0.0000	0.0000
5	7	0.0525	-0.0005	0.0012	0.0000	0.0000	-0.0001
	8	0.0244	-0.0004	0.0014	0.0000	0.0000	-0.0001
	9	0.0502	-0.0027	-0.0134	-0.0001	0.0001	-0.0001
	10	0.0519	-0.0005	0.0012	0.0000	0.0000	-0.0001
	11	0.0521	-0.0006	0.0007	0.0000	0.0000	-0.0001
6	7	0.0147	-0.0014	0.0012	0.0000	0.0000	0.0000
	8	-0.0178	-0.0015	0.0014	0.0000	0.0000	0.0000
	9	0.0158	-0.0002	-0.0136	-0.0001	0.0001	0.0000
	10	0.0134	-0.0014	0.0012	0.0000	0.0000	0.0000
	11	0.0149	-0.0013	0.0006	0.0000	0.0000	0.0000
7	7	0.0745	-0.0008	0.0022	0.0000	0.0000	-0.0001
	8	0.0487	-0.0007	0.0026	0.0000	0.0000	-0.0001
	9	0.0719	-0.0035	-0.0206	0.0000	0.0001	-0.0001
	10	0.0736	-0.0008	0.0023	0.0000	0.0000	-0.0001
	11	0.0741	-0.0010	0.0013	0.0000	0.0000	-0.0001
8	7	0.0221	-0.0020	0.0022	0.0000	0.0000	0.0000
	8	-0.0097	-0.0019	0.0026	0.0000	0.0000	-0.0001
	9	0.0232	-0.0006	-0.0207	0.0000	0.0001	0.0000
	10	0.0203	-0.0020	0.0023	0.0000	0.0000	0.0000
	11	0.0222	-0.0019	0.0012	0.0000	0.0000	0.0000
9	7	0.0956	-0.0012	0.0033	0.0000	0.0000	-0.0001
	8	0.0780	-0.0010	0.0039	0.0000	0.0000	-0.0002
	9	0.0932	-0.0040	-0.0264	0.0000	0.0001	-0.0002
	10	0.0946	-0.0012	0.0034	0.0000	0.0000	-0.0001
	11	0.0954	-0.0013	0.0019	0.0000	0.0000	-0.0001
10	7	0.0302	-0.0024	0.0033	0.0000	0.0000	-0.0001
	8	0.0050	-0.0023	0.0039	0.0000	0.0000	-0.0001
	9	0.0311	-0.0010	-0.0264	0.0000	0.0001	-0.0001

JOINT DISPLACEMENT (CM   RADIANS)   STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN